

HALLI, Finland

WMO: 029450

Lat: 61.855N

Lon: 24.805E

Elev: 145

StdP: 99.60

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.1	-22.7	-28.8	0.3	-26.2	-24.9	0.4	-22.7	9.4	0.6	8.1	-0.4	1.2	320	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	26.9	17.7	25.0	16.9	23.1	16.0	19.3	24.2	18.2	22.8	17.2	21.4	3.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	13.0	20.8	16.5	12.0	19.6	15.5	11.2	18.8	55.5	23.9	52.0	22.7	48.8	21.3	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.8	6.0	-29.5	28.9	4.2	2.0	-32.5	30.3	-34.9	31.4	-37.2	32.6	-40.2	34.0		
(5)			-29.4	20.6	4.1	1.5	-32.4	21.6	-34.8	22.5	-37.1	23.3	-40.1	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-7.3	-7.4	-3.3	3.1	9.2	13.9	16.9	15.2	10.1	4.2	-0.4	-4.4		
	DBStd	9.71	7.04	7.10	4.89	3.66	4.35	3.58	3.24	3.15	3.47	4.03	4.48	6.59		
	HDD10.0	2701	536	488	411	209	68	6	0	2	42	181	312	447		
	HDD18.3	5192	794	721	669	457	285	140	68	106	248	437	562	705		
	CDD10.0	589	0	0	0	2	43	122	214	163	44	2	0	0		
	CDD18.3	39	0	0	0	0	1	7	23	8	0	0	0	0		
	CDH23.3	359	0	0	0	0	20	71	206	63	0	0	0	0		
	CDH26.7	54	0	0	0	0	2	8	39	6	0	0	0	0		
(14) Wind	WSAvg	2.7	2.6	2.6	2.9	2.7	2.9	2.9	2.5	2.5	2.6	2.8	2.8	2.9		
(15) Precipitation	PrecAvg	664	54	38	36	32	38	68	90	75	54	68	55	51		
	PrecMax	898	91	75	69	56	94	114	146	157	134	131	125	93		
	PrecMin	564	18	2	8	7	6	28	18	24	12	13	6	16		
	PrecStd	72	21	20	14	15	21	25	32	38	31	32	30	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	4.6	9.9	18.8	25.8	27.8	29.7	27.2	21.1	13.9	9.3	6.9		
		MCWB	3.4	2.5	4.6	10.7	15.1	17.6	19.3	18.0	15.6	11.4	8.7	6.2		
	2%	DB	2.5	2.8	6.6	14.9	22.7	25.4	27.7	25.0	18.9	11.9	7.8	5.1		
		MCWB	2.1	2.0	3.0	8.3	13.7	16.1	18.7	17.5	14.6	10.5	7.2	4.8		
	5%	DB	1.5	1.6	4.4	12.2	20.0	23.1	25.7	23.3	17.0	11.0	6.7	3.8		
		MCWB	1.2	1.1	1.9	6.7	12.3	15.2	17.7	16.5	13.5	9.9	6.3	3.4		
	10%	DB	0.7	0.6	2.8	10.0	17.4	21.1	23.7	21.4	15.3	9.9	5.2	2.5		
		MCWB	0.4	0.2	1.1	5.6	10.9	14.2	17.0	15.9	12.8	9.0	4.8	2.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	3.1	5.4	11.4	16.5	19.3	21.1	20.0	16.3	12.2	8.8	6.2		
		MCDB	3.3	3.8	8.7	15.6	22.9	24.9	26.1	24.7	19.2	13.2	9.2	6.5		
	2%	WB	2.1	2.0	3.5	9.2	14.8	17.6	19.9	18.5	15.2	11.0	7.2	4.8		
		MCDB	2.3	2.7	5.6	13.4	20.1	22.6	25.0	22.6	17.5	11.6	7.5	5.0		
	5%	WB	1.2	1.2	2.3	7.5	13.3	16.3	18.8	17.5	14.3	10.1	6.1	3.3		
		MCDB	1.4	1.6	3.9	11.6	18.7	21.1	23.6	21.4	16.4	10.8	6.5	3.6		
	10%	WB	0.3	0.2	1.3	5.9	11.6	15.1	17.8	16.6	13.2	9.0	4.9	2.1		
		MCDB	0.5	0.5	2.4	9.3	16.5	19.9	22.4	20.2	15.0	9.6	5.2	2.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.3	8.0	9.1	10.3	9.5	9.3	9.2	7.6	5.0	3.8	4.6		
		MCDBR	4.1	4.8	9.3	13.6	14.8	13.1	12.4	12.1	9.5	5.5	4.2	3.9		
	5% WB	MCWBR	4.0	4.5	6.5	8.0	7.1	5.8	5.0	5.4	5.7	4.4	4.2	3.9		
		MCWBR	3.9	4.5	7.7	12.1	12.7	10.9	10.5	10.0	8.0	5.0	4.1	3.8		
(40) Clear-Sky Solar Irradiance	taub		0.242	0.289	0.317	0.351	0.349	0.346	0.369	0.355	0.332	0.316	0.264	0.356		
		taud	2.063	2.153	2.231	2.374	2.455	2.487	2.438	2.470	2.527	2.417	2.162	2.614		
	Ebn at Noon		442	664	789	839	873	881	846	830	786	649	415	287		
		Edh at Noon	37	68	91	97	97	96	99	88	70	54	33	22		
(44) All-Sky Solar Radiation	RadAvg		0.21	0.77	2.24	3.66	4.92	5.21	5.01	3.89	2.25	0.94	0.25	0.11		
	RadStd		0.04	0.18	0.27	0.41	0.55	0.41	0.51	0.45	0.29	0.17	0.05	0.01		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (54 neighbors)		+0.50	N/A	N/A	N/A	N/A	N/A	-141	N/A	N/A	N/A	

Nomenclature: See separate page